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Question **10**

Not yet answered

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Flag question

The problems of fluctuating loads are overcome by

a. **Using chopper to control the motor speed**

b. **Using a gearbox between the motor and the load**

c. **Using coupling element between the motor and the load**

d. **Mounting a flywheel on the motor shaft in non-reversit**

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/ General / Quiz 1

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Question 7

Not yet answered

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Flag question

The hardness of series field dc motors torque-speed characteristic is

a. Zero

b. Negative

c. Infinity

d. Positive

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Question 4

Not yet answered

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Rotating mechanical load of 10 N.m connected to the motor through gearbox , which has gear ratio of 5. If the system efficiency equals 0.9, the referred load torque to motor shaft is

- a. **50 N.m**
- b. **2.22 N.m**
- c. **2 N.m**
- d. **1.8 N.m**

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Jump to...

week-1 : Electric drive system →

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Question **9**

Not yet answered

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Flag question

In paper rewinding machine

- a. Torque is inversely proportional to speed
- b. Torque is proportional to speed square
- c. Torque is directly proportional to the speed
- d. Torque is independent of the speed

Clear my choice

CamScanner الممسوحة ضوئيا بـ

The coefficient B in the equation $T = T_L + J\frac{d^2\theta}{dt^2} + B\omega$, represents

- a. [illegible]
- b. updated viscous friction
- c. [illegible]
- d. [illegible]

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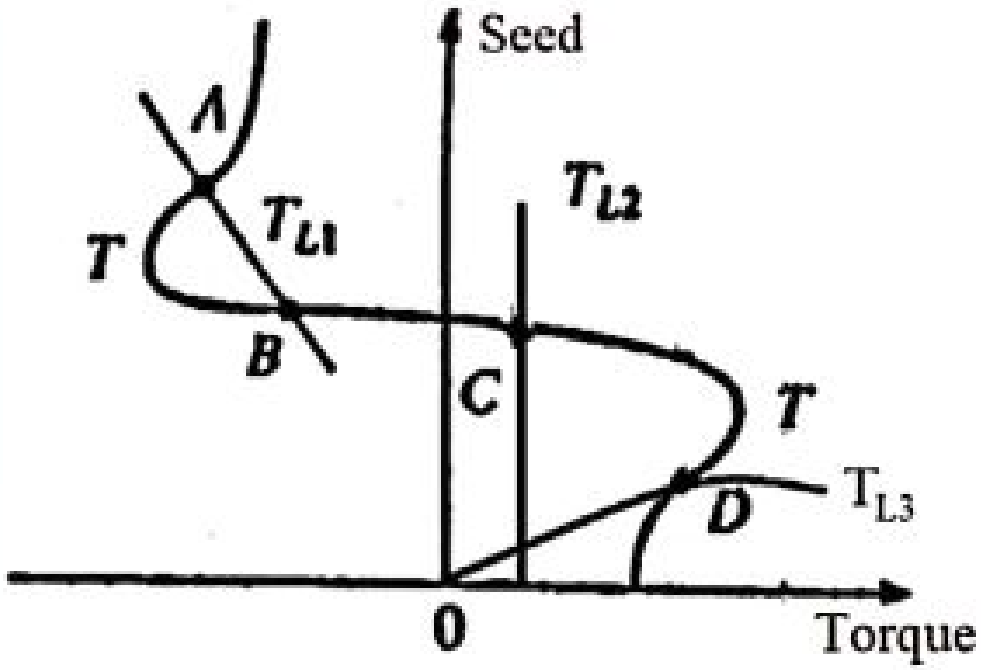
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Flag question

stabilities for points A,B,C,D respectively in the figure below are



- a. unstable, unstable, stable, unstable
- b. unstable, stable, stable, unstable
- c. unstable, stable, stable, stable
- d. unstable, unstable, stable, unstable

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General / Quiz 1
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The hardness of series field dc motors torque-speed characteristic is

a. Zero

b. Negative

c. Infinity

d. Positive

CamScanner الممسوحة ضوئيا بـ

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lag question

e equivalent moment of inertia referred to motor shaft calculation is based the

a. Heating (electric) power conservation law

b. Kinetic energy conservation law

c. Mechanical power conservation law

d. Electro-magnetic energy conservation law

Clear my choice

الممسوحة ضوئيا بـ CamScanner

i23-1-ELE2461-ELECTRICAL DRIVES-1

Referring to the drive systems, Lathe cutting machine is considered

- a. inductive load
- b. passive load
- c. active load
- d. resistive load

Clear my choice

Jump to...

a. active load

b. resistive load

c. passive load

d. inductive load

Clear my choice

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A drive has the following parameters:

J=10 kg.m2, T=15+0.05N, TL=5+0.06N, where N is speed in RPM

Initially, the drive is working in steady-state, Now the drive is braked by electrical braking.

Torque of the motor in braking is T=10-0.04N, calculate the time taken by the drive to fully stop(zero).

- a. 25.2 S
- b. 20.3
- c. 22 S
- d. 21.38 S

Clear my choice

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Jump to...

الممسوحة ضوئيا بـ CamScanner

In paper rewinding machine

- a. Torque is independent of the speed
- b. Torque is directly proportional to the speed
- c. Torque is inversely proportional to speed
- d. Torque is proportional to speed square

ious page

Search

CamScanner الممسوحة ضوئيا بـ

'2023-1-ELE2461-ELECTRICAL DRIVES-1

The power factor of the dc motor when it operates at steady-state point is

- a. 1
- b. $V_a E_a / V_a I_a$
- c. $P_d / V_a I_a$
- d. $I_a E_a / V_a I_a$

Clear my choice

الممسوحة ضوئيا بـ CamScanner

23 - EEE 107 ELECTRICAL DRIVES-1

A drive has the following parameters:

J=10 kg.m2, T=15+0.85NL TL=5+0.05N, where N is speed in RPM

Initially, the drive is working in steady-state. Now the drive is braked by electrical braking.

Torque of the motor in braking is T=10-0.04N, calculate the time taken by the drive to fully stop(zero).

- a. 25.25
- b. 225
- c. 21.38 S
- d. 263

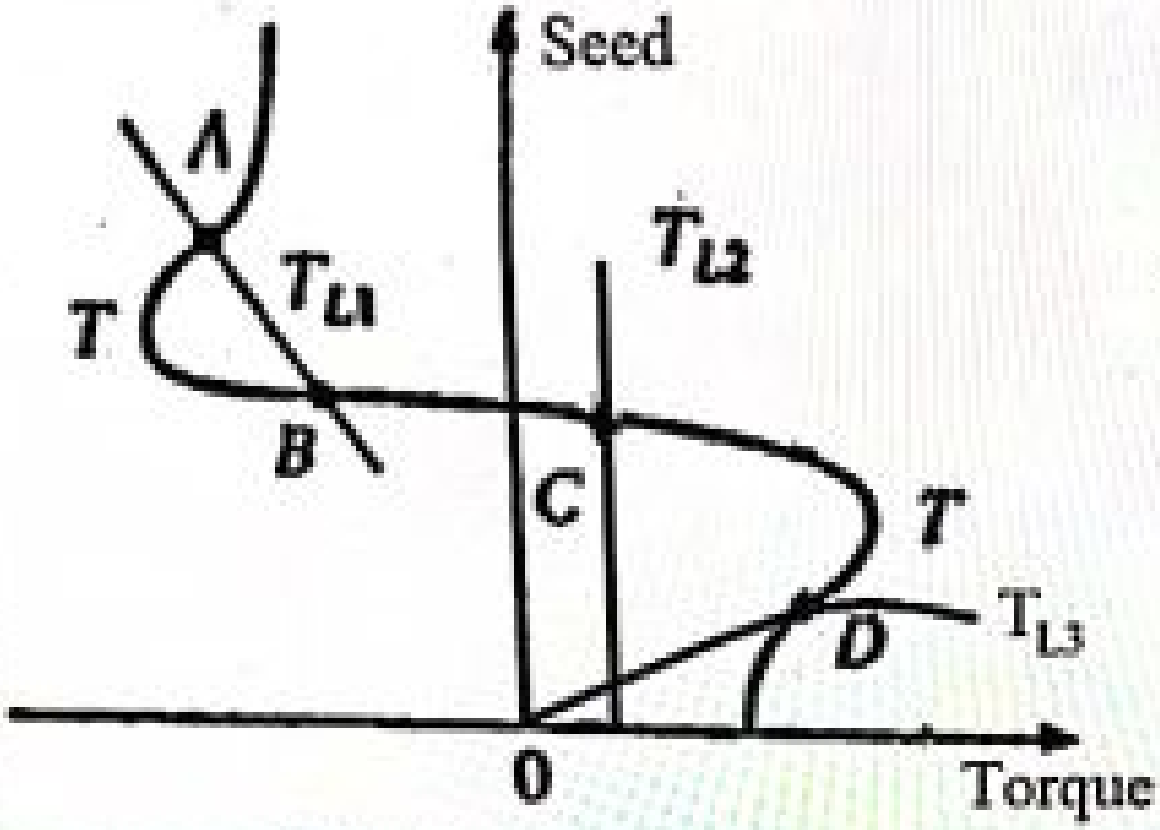
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stabilities for points A,B,C,D respectively in the figure below are



- a. unstable, stable, stable, unstable
- b. unstable, unstable, stable, unstable
- c. unstable, unstable, stable, unstable
- d. unstable, stable, stable, stable

Clear my choice

الممسوحة ضوئيا بـ CamScanner

2023-1-ELE2461-ELECTRICAL DRIVES-1

If motor rotational speed equals 1450 rpm, the angular speed is

- a. 152 rad/s
- b. 150 rad/s
- c. 461 rad/s
- d. 1225 289.42 rad/s

Clear my choice

الممسوحة ضوئيا بـ CamScanner

The drive system output power is:

- a. **Electrical power**
- b. **Thermal power**
- c. **Mechanical power**
- d. **Electromagnetic power**

Clear my choice

CamScanner الممسوحة ضوئيا بـ

-1-ELE2461-ELECTRICAL DRIVES-1

Time left 0:1

Electric motor has moment of inertia of 1kg.m^2 directly connected to motor shaft drum with moment of inertia 4 kg.m^2. the drive system total (equivalent) moment of inertia equals:

- a. 5 kg.m^2
- b. 1 kg.m^2
- c. 4 kg.m^2
- d. 3 kg.m^2

Clear my choice

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Jump to...

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The coefficient B in the equation $T = T_L + J\frac{d\omega}{dt} + B\omega$, represents

- a. viscous friction
- b. updated viscous friction
- c. safety factor
- d. dry friction

Clear my choice

CamScanner الممسوحة ضوئيا بـ

$T = T_L$

N_1 :

$15 + 0.05N = 5 + 0.06N$

$10 = 0.01N$, $N = 1000$ rpm

N_2 = zero

During Braking $T = 10 - 0.04N$, $T_L = 5 + 0.06N$

$$J\frac{dw}{dt} = T - T_L$$

$$10 * \frac{2\pi}{60} * \frac{dN}{dt} = 10 - 0.04N - 5 - 0.06N$$

$$\frac{\pi}{3} * \frac{dN}{dt} = 5 - 0.1N$$

$$dt = \frac{3}{\pi}\int_{1000}^{0} \frac{dN}{5 - 0.1N}$$ ≈ Divergence

$T_2 = -10 - 0.04N$

$$\frac{\pi}{3} * \frac{dN}{dt} = -15 - 0.1N$$

$$t = \frac{3}{\pi}\int_{1000}^{0} \frac{dN}{-15 - 0.1N} \quad , t = 21.33$$

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-1-ELE2461-ELECTRICAL DRIVES-1

Time left 0:03:30

A drive system has following parameter:

$J = 0.25 kg.m^2$, $T_L = 3$ N.m .

Initially the drive system is working in steady state. The steady-state speed is1371 rpm. Now the drive is disconnected from the main supply. Calculate time taken by the drive to stop (consider that the stopping process is finished at 0.05 of final value).

- a. 91.2s
- b. 23.4 s
- c. 5.85s
- d. 15.6s

الممسوحة ضوئيا بـ CamScanner

Palestine Technical University
Faculty Of Engineering Technology
Mechatronics Engineering Dept.
Fundamentals of Elec. Drive

Time of Exam: 80 Min.
12/09/2022

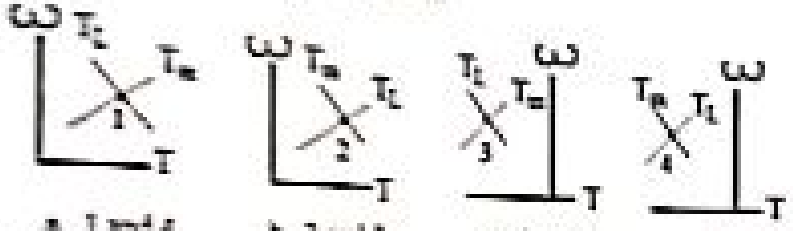
Final exam
50 Points

Prof. Dr Mahdi Abbaraqun	Instructor

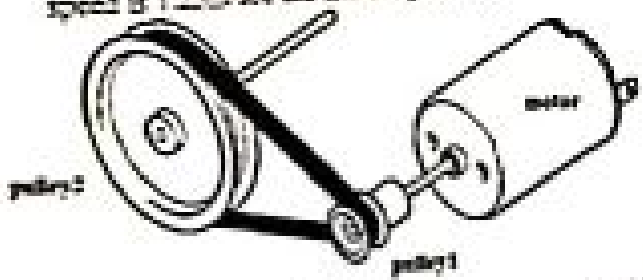
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Q1: (25 marks)
Answer the following questions by filling the table at the end of this question

1. The parallel connection of ac motor stator windings will cause variation of:
   a. Stator voltage b. No-load speed
   c. Load torque d. Stator frequency
2. Unstable points of equilibrium are



   a. 1 and 4 b. 2 and 3 c. 1 and 3 d. 2 and 4
3. Conveyor torque –speed characteristic is:
   a. Constant torque characteristic b. Linearly rising torque characteristic
   c. Non-Linearly rising torque characteristic d. Non-Linearly decreasing torque characteristic
4. If the torque developed by the motor is $T = T_L \mp J\frac{d\omega}{dt}$, the motor is working at
   a. no-load condition b. transient condition c. S.S.P d. none of these
5. The motion equation $T = T_L + J\frac{d\omega}{dt} = S\omega$ can be solved analytically if
   a. T and T_L are constants b. T or T_L is neither constant nor proportional to speed
   c. T and T_L are constants or proportional to speed d. none of these.
6. The equivalent moment of inertia referred to motor shaft calculated depending on:
   a. Heating (electric) power conservation law b. Mechanical power conservation law
   c. Electro-magnetic energy conservation law d. Kinetic energy conservation law
7. In the figure below, diameters of pulley 1 and pulley2 are 10cm and 50cm respectively, if the belt speed is 1.25m/sec the motor speed is



   a. 240rpm b. 250rpm c. 12rpm d. none of these
8. One of the following dc motors does not operate in regenerating mode:
   a. Parallel field dc motor b. Series field dc motor
   c. Compound field dc motor d. Separately excited dc motor
9. When the motor speed is controlling by stator voltage changing, the permissible torque is governed by
   a. $T_{per} = T_s S/S_s$ b. $T_{per} = T_s S_s/S$ c. $T_{per} = T_s S_s/S$ d. $T_{per} = T_s S_{cr}/S$

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d. all of these

Component determines the motor starting time is
a. motor inertia b. motor load c. speed change rate

$$\omega = \frac{3V_1^2 R_2}{\left[R_1 + \sqrt{R_1^2 + (X_{1s} + X_2')^2}\right]}$$ determines:

1. The following expression
Maximum Torque b. Motor no-load torque
Starting torque d. Motoring mode torque

12. If the variable voltage variable frequency method is used and the load is ventilator, the low which governs the motor operation is
a. $\frac{V}{\sqrt{f}}$ =constant b. $\frac{V}{f^2}$ =constant c. $\frac{V}{f}$ =constant d. none of these

13. If the line voltage and frequency are V_L=380V, f=50Hz, then the most suitable motor to be started by Y/A is
a. Y/A 660/1143V b. Y/A 660/380V
c. Y/A 380/220V d. Y/A 220/110V

14. Braking mode that is used to store reactive energy:
a. dynamic b. zero-sequence braking c. plugging d. regenerative

15. A Y - connected three phase motor consumes 10A from the line, if it is reconnected by Δ for the same line voltage it will consume current equals to
a. 30A b. 10A c. 17A d. 50A

16. If the synchronous motor load torque value is more than synchronizing torque during starting, the motor will
a. Run in short circuit mode
b. Not be synchronized and operate as synchronous motor
c. Be synchronized and operate as synchronous motor
d. Be synchronized and operate as synchronous generator

17. If the nominal speed of squirrel- cage motor is 980 rpm, the motor has
a. 2 poles b. 4 poles c. 6 poles d. none of these

$$P_{eq} = \sqrt{\frac{\sum_{i=1}^{n} P_i^2 t_i}{\sum_{i=1}^{n} t_i}}$$

18. The relationship used to calculate equivalent load power of
a. Continuous duty cycle with constant load b. Continuous duty cycle with fluctuating periodic loading
c. Short time duty cycle d. Continuous duty cycle with intermittent periodic loading

19. Indirect starting of electric motor is used when:
a. The motor is medium and high power and the source is limited power supply
b. The motor is fractional power and the source is unlimited power supply
c. The motor is medium and high power and the source is unlimited power supply
d. none of these

20. The dc motors allowed starting current value is
a. (2-3) times of nominal current b. (5-7) times of nominal current
c. Nominal current d. Less than nominal current

21. The ac motor power rating is
a. proportional to the motor nominal slip b. not depending on the motor nominal slip
c. inversely proportional to the motor nominal slip d. none of these

22. In the intermittent periodic duty, the motor temperature can
a. reach the equilibrium temperature at rest period
b. reach the equilibrium temperature at first period

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مبيضين · 59m

REPLY MARK AS READ

[redacted]

Ans→ Stator resistance, $R_1 = 3\Omega$, Stator reactance, $X_1 = 0.5X_{sc}$

Rotor resistance, referred to stator $\boxed{R_2 = 2\Omega}$ $\therefore X_1 = 0.5 \times 10 = 5\Omega$

Rotor reactance, $X_2 = 0.5X_{sc} = 0.5 \times 10 = 5\Omega$ $\therefore \boxed{X_2 = 5\Omega}$

Torque in an Inductor motor is given by :-

$$T_{em} = \frac{3 \times 60}{2\pi N_s} \frac{sV^2 R_2}{(R_2^2 + sX_2^2)} \quad , \quad s = \frac{N_s - N}{N_s} \quad , \quad N_s \rightarrow \text{synchronous speed}$$

$$N_s = \frac{120f}{P}$$

$V \rightarrow$ Voltage applied to the Induction motor (Phase)

At starting speed, $N = 0$ $\therefore s = \frac{N_s - 0}{N_s} = 1$

$\therefore$ At starting, Torque, $T_{em} = \frac{3 \times 60}{2\pi N_s} \cdot \frac{V^2 R_2}{(R_2^2 + X_2^2)}$

$\therefore T_{em} \propto V^2$ $\therefore \frac{T_{em_1}}{T_{em_2}} = \frac{V_1^2}{V_2^2}$

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1] ?

2] c

3] a

4] B

5] ?

6] d

7] a

8] b

9] c

10] d

11] c

12] ?

13] c

14] ?

15] c

16] B

17] c

18] d

19] a

20] a

21] ?

22] ?

23] c

24] c

25] ? a

26] ?

27] d

28] ?

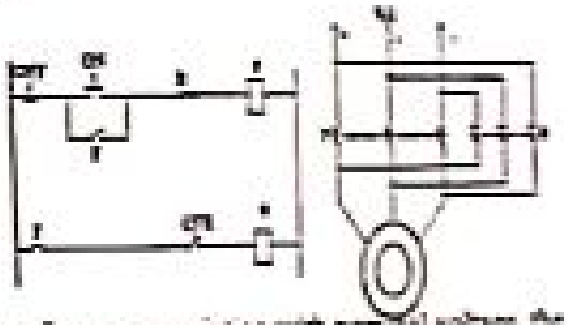
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c. reach the equilibrium temperature at the on-time interval
d. reach the equilibrium temperature at many periods

23. If mechanical time-constant related to electro-magnetic time-constant of drive system value is less than four, then drive system characteristic equation has:
a. Two different real roots
b. Two equal real roots
c. Two different complex roots
d. Two equal complex roots

24. Centrifugal relay is used for automatic starting and braking of electric motors according to:
a. Position principle
b. Voltage principle
c. Speed Principle
d. Time principle

25. The given electrical diagram shows
a. direct starting and mechanical braking
b. Starting in forward direction and plugging braking
c. Starting in forward direction and dynamic braking
d. Forward working and backward working



26. In speed regulation of three-phase ac motors by stator frequency variation with nominal voltage, the frequency should not be less than the nominal value to protect the motor from
a. Overheating
b. Under loading
c. Over voltage
d. Over loading

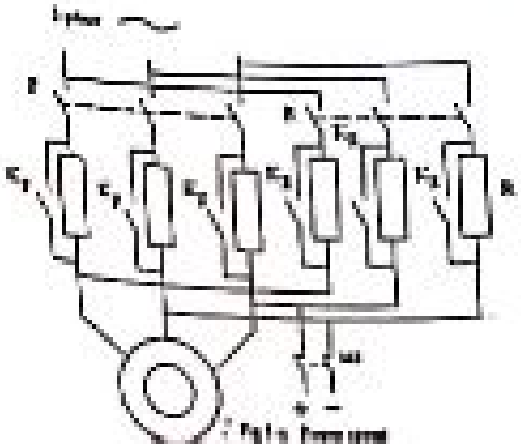
27. According to the A/C motor speed control by varying stator voltage with fixed frequency, Load torque should be subjected to have constant
a. Load torque
b. Critical torque
c. Starting torque
d. Motor loading capacity

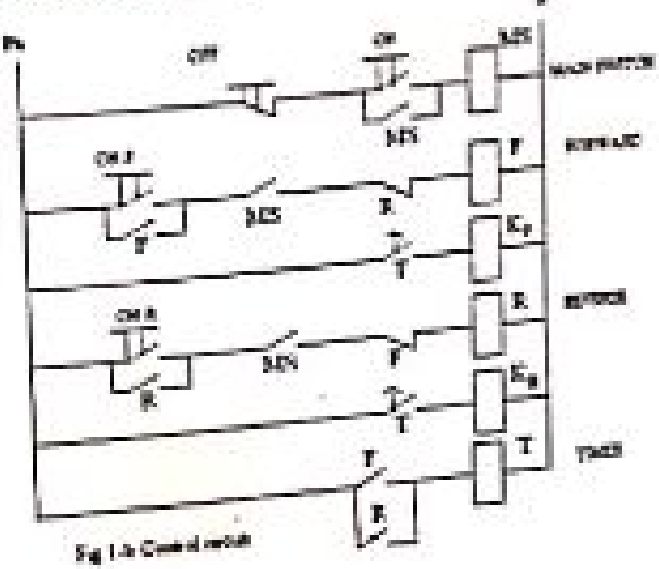
28. Increasing the input supply frequency of three-phase ac motor more than nominal frequency with fixed stator voltage will lead to
a. Decrease stator field angular speed
b. Increase motor starting torque
c. Decrease field flux
d. Increase motor load torque

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Ans														

Q	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Ans														

Q2: Given the circuit diagrams of control and power for squirrel-cage motor below (8 marks)





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Since $V_2 = 0.9V_1$ (10% decrease in voltage)

$$\therefore \frac{T_{em1}}{T_{em2}} = \frac{V_1^2}{(0.9V)^2} = \frac{1}{0.81} \Rightarrow \frac{T_{em1}}{T_{em2}} = \frac{1}{0.81}$$

$$\therefore \%\text{Change in starting Torque} = \frac{(T_{em1} - T_{em2})}{T_{em1}} \times 100$$

$$= \left(\frac{T_{em1}}{T_{em1}} - \frac{T_{em2}}{T_{em1}}\right) \times 100 = \left[1 - \frac{0.81}{1}\right] * \times 100 = 19\%$$

$\therefore$ **19% change in starting Torque if voltage is reduced by 10%**

Condition for Maximum Torque, put $\frac{d(T_{em})}{ds} = 0$

$$\therefore \frac{d}{ds}\left[\frac{3 \times 60 \, s V^2 R_2}{2\pi N_s (R_2^2 + s^2 X_2^2)}\right] = 0 \Rightarrow \boxed{sX_2 = R_2}$$

$\therefore$ Condition for Maximum Torque, $\boxed{sX_2 = R_2}$

Since at starting slip, $\boxed{s = 1}$ (Calculated in previous part)

$\therefore$ for Maximum Torque at starting, $\boxed{X_2 = R_2}$

$\therefore$ **Rotor reactance = Rotor resistance.** $\rightarrow$ for Maximum starting.

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Let R_{ex} is resistance added to rotor circuit, therefore rotor resistance, $R_2' = R_2 + R_{ex}$

$\therefore$ $X_2 = R_2'$ $\Rightarrow$ $R_2 + R_{ex} = X_2$

$\Rightarrow$ $2 + R_{ex} = 5$ $\Rightarrow$ $R_{ex} = 5 - 2 = 3\Omega$

$\therefore$ $\boxed{R_{ex} = 3\Omega}$

$\therefore$ 3Ω resistance should be added to rotor circuit to achieve Maximum torque at starting

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Draw the static mechanical characteristics

Draw the dynamic characteristics for speed and Torque

(7 marks)

duction motor has a stator resistance of 1Ω and a rotor resistance referred to the stator is 2Ω, the
al inductive reactance X_{sc} =10Ω, calculate

a. The change in starting torque if the stator voltage is reduced by 20%
b. The resistance that should be added to the rotor circuit to achieve the maximum torque at starting

Solution



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For 3-phase full-wave converter drive, the converter in the field circuit of sep.excited dc motor should be

- a. single- or three-phase full converter
- b. dual converter
- c. single- phase full converter
- d. 3-phase converter

الممسوحة ضوئيا بـ CamScanner

/2023-1-ELE2461-ELECTRICAL DRIVE

Nominal slip (S_n) of induction motors

- a. is directly proportional to the motor power
- b. doesn't depend on the motor power
- c. is inversely proportional to the motor power
- d. is directly proportional to the square of motor power

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2023-1-ELE2461-ELECTRICAL DRIVES-1

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The method that is not adequate for motor selection when the motor speed is changing considerably under load and at starting and braking conditions

- a. equivalent torque
- b. equivalent losses
- c. equivalent current
- d. equivalent power

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2024/2023-1-ELE2461-ELECTRICAL DRIVES-1

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Time left 0:13:53

Question 4

Not yet answered

Marked out of 1.00

Flag question

Standard CDFs are

- a. 15, 25, 40 and 60%
- b. 10, 25, 40 and 60%
- c. 15, 25, 35 and 60%
- d. 15, 25, 40 and 65%

Clear my choice

Quiz na

1 2

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الممسوحة ضوئيا بـ CamScanner

With field reversal, the three-phase full-wave-converter drive is

- a. four-quadrant drive
- b. three quadrant drive
- c. one quadrant drive
- d. two-quadrant drive

Clear my choice

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quiz 2 (page 3 of 13)

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For 3-phase full wave converter drive, the thyristor gate- pulse duration is

a. 30^0

b. 60^0

c. 180^0

d. 120^0

Clear my choice

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quiz 2 (page 1 of 15)

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Time left 0:16:55

Question 1

Not yet answered

Marked out of 1.00

Flag question

For single-phase full converter drive of sep.excited dc motor, the reversal of the armature or field allows operation in the

- a. second and third quadrants
- b. first and second quadrant
- c. third and fourth quadrant
- d. first and third quadrant

Clear my choice

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24/2023-1-ELE2461-ELECTRICAL DRIVES-

Time

6

For dc motors, the converter in the field circuit could be

- a. semi or full
- b. full only
- c. semi or full or dual
- d. semi only

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Time left 0:15:55

For 3-phase full-wave converter drive, the converter in the field circuit of sep.excited dc motor should be

- a. single- or three-phase full converter
- b. single- phase full converter
- c. dual converter
- d. 3-phase converter

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2.chalqa.comsats/mdlmoodle/mod/quiz/attempt.php?attempt=376528&cmid=265338&page=1

)23-1-ELE2461-ELECTRICAL DRIVES-1

Time left 0:15:3

The average output of a single-phase semiconverter connected to 120V, 50Hz,supply and firing angle of π/2 is

- a. 120V
- b. 54.02V
- c. 27V
- d. 0V

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quiz 2 (page 3 of 15)

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2023-1-ELE2461-ELECTRICAL DRIVES-1

urses / 2024/2023-1-ELE2461-ELECTRICAL DRIVES-1 / General / quiz 2

Time left 0:14:22

For single-phase full converter drive of sep.excited dc motor, the reversal of the armature or field allows operation in the

- a. first and second quadrant
- b. first and third quadrant
- c. third and fourth quadrant
- d. second and third quadrants

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2023-1-ELE2461-ELECTRICAL DRIVES-1

Time left 0:11:3

A motor has a heating time constant of 60s and cooling time constant of 90s, if the motor rating is 10kw, find the permitted load power that can be used for 10s to let the motor work at its optimal utilization.

- a. 5!kw
- b. 1.275kw
- c. 2.55kw
- d. 25.5kw

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Rotor copper losses in induction motors are equal to

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a. $T\omega_s$

b. $T\omega_s / s$

c. $sT_s\omega_s$

d. $sT\omega_s$

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If we change the terminal voltage (stator voltage) of induction motor,

- a. we must keep the stator flux constant
- b. we must keep the motor load capacity constant ($\lambda = \lambda_n$)
- c. we must keep the load constant (load equals the nominal load)
- d. we must keep the rotor losses constant (equal the losses at motor ratings)

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Use of a motor having insufficient rating leads to:

- a. operate below rated power factor especially IMs & synchronous motors
- b. oversizing of other components in the system, impacting the overall design and efficiency
- c. electrical instabilities and power quality issues
- d. increase the productivity

Clear my choice

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